

QIXIAN SHAN, China

WMO: 587260

Lat: 27.960N

Lon: 117.841E

Elev: 1415

StdP: 85.44

Time Zone: 8.00 (E08)

Period: 82-91

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.4	-5.8	-20.5	0.7	1.8	-16.8	1.0	1.6	14.9	4.9	12.7	4.6	2.7	320	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.3	26.0	21.4	25.0	21.1	24.1	20.6	22.1	24.9	21.6	24.2	21.1	23.6	1.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	18.9	24.0	20.7	18.3	23.5	20.2	17.8	22.8	72.9	25.1	70.8	24.3	68.5	23.7	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	11.4	9.3	-10.2	28.9	1.4	1.9	-11.2	30.3	-12.0	31.4	-12.8	32.5	-13.9	33.8		
			-10.9	23.6	1.4	1.3	-11.9	24.5	-12.7	25.2	-13.5	25.9	-14.5	26.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	2.5	3.3	7.0	11.7	15.8	19.0	21.9	20.8	17.6	13.6	8.8	4.4	(6)
(7)		DBStd	7.67	4.45	4.76	5.05	4.16	3.42	2.56	1.92	1.79	2.73	3.41	4.29	5.07	(7)
(8)		HDD10.0	837	235	193	120	30	3	0	0	0	8	69	178		(8)
(9)		HDD18.3	2498	492	422	352	201	89	22	2	3	46	153	285	431	(9)
(10)		CDD10.0	1660	2	4	27	80	183	272	370	336	227	118	35	6	(10)
(11)		CDD18.3	278	0	0	1	1	12	43	114	80	23	4	1	0	(11)
(12)		CDH23.3	378	0	0	1	0	8	36	233	90	9	1	0	0	(12)
(13)		CDH26.7	12	0	0	0	0	1	1	7	3	1	0	0	0	(13)
(14)	Wind	WSAvg	3.4	3.3	3.7	3.9	4.1	3.7	3.6	3.6	3.1	2.9	2.8	3.1	2.9	(14)
(15)	Precipitation	PrecAvg	1932	86	108	213	230	280	368	164	177	94	59	90	64	(15)
(16)		PrecMax	2526	224	225	443	418	558	802	393	330	250	212	248	235	(16)
(17)		PrecMin	1211	8	12	87	110	108	115	19	71	29	8	9	5	(17)
(18)		PrecStd	403	49	50	81	82	110	156	89	76	48	52	70	60	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	15.4	19.1	22.1	24.3	25.9	27.6	26.5	24.7	23.1	20.1	14.8	(19)
(20)		MCWB	9.3	11.2	15.1	17.6	19.5	21.0	21.9	21.4	21.1	19.9	15.8	8.3		(20)
(21)		2%	DB	11.4	13.0	16.6	19.9	22.4	24.3	26.4	25.2	23.0	21.0	17.4	13.3	(21)
(22)		MCWB	8.0	10.3	14.1	16.4	18.2	20.7	21.7	21.2	20.0	18.4	13.9	8.0		(22)
(23)		5%	DB	9.9	11.5	15.1	18.6	21.3	23.3	25.5	24.4	22.0	19.5	15.7	11.9	(23)
(24)		MCWB	7.7	9.7	13.2	15.4	17.9	20.3	21.4	20.9	19.4	16.8	12.3	7.4		(24)
(25)		10%	DB	8.8	10.3	14.0	17.2	20.2	22.3	24.7	23.5	21.0	18.0	14.3	10.8	(25)
(26)		MCWB	6.2	8.9	12.4	14.2	17.3	19.5	21.0	20.4	18.7	15.8	11.3	6.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	12.0	15.6	18.5	20.2	22.1	23.1	22.6	21.6	20.4	16.6	10.9	(27)
(28)		MADB	12.2	13.8	18.8	21.4	22.7	24.2	26.3	25.0	23.8	22.5	19.2	12.6		(28)
(29)		2%	WB	9.4	11.0	14.4	17.2	19.5	21.4	22.4	21.8	20.6	18.9	15.1	9.8	(29)
(30)		MADB	10.6	12.5	16.3	19.2	21.5	23.5	25.4	24.2	22.3	20.5	16.9	11.8		(30)
(31)		5%	WB	8.5	10.1	13.5	16.1	18.7	20.7	21.9	21.3	19.9	17.5	13.6	8.9	(31)
(32)		MADB	9.5	11.3	14.9	17.9	20.5	22.7	24.8	23.6	21.4	19.0	15.0	11.0		(32)
(33)		10%	WB	7.2	9.2	12.5	15.1	18.0	20.1	21.3	20.8	19.2	16.4	12.4	7.7	(33)
(34)		MADB	8.2	10.1	13.7	16.8	19.8	21.8	24.2	23.0	20.5	17.6	13.6	10.0		(34)
(35)	Mean Daily Temperature Range		MDBR	5.3	5.0	5.2	4.8	4.5	4.0	4.3	4.3	3.9	4.5	5.2	5.7	(35)
(36)		5% DB	MADB	6.0	6.0	5.7	5.8	5.5	5.0	5.5	5.3	4.8	5.7	6.4	6.3	(36)
(37)			MCWBR	4.8	4.4	4.2	4.7	4.2	4.0	4.3	4.0	4.6	5.3	4.9	(37)	
(38)			MADB	5.8	5.6	5.4	5.1	4.8	4.4	4.9	4.7	4.2	4.8	5.6	5.8	(38)
(39)		5% WB	MCWBR	4.6	4.4	4.1	4.1	3.8	3.7	4.4	3.8	3.6	3.8	4.4	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.511	0.550	0.610	0.595	0.531	0.518	0.485	0.522	0.517	0.516	0.477	0.465	(40)	
(41)		taud	1.770	1.720	1.630	1.714	1.926	1.998	2.116	1.990	1.951	1.853	1.900	1.887	(41)	
(42)		Ebn at Noon	698	714	702	733	782	787	813	781	770	735	729	722	(42)	
(43)		Edh at Noon	192	218	253	239	194	179	159	179	181	188	166	163	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.17	2.47	3.03	3.56	3.98	3.93	5.47	4.84	4.20	3.64	2.68	2.44	(44)	
(45)		RadStd	0.46	0.61	0.38	0.51	0.59	0.55	0.52	0.45	0.35	0.58	0.44	0.50	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (8 neighbors)	+0.30	N/A	N/A	+0.44	N/A	N/A	N/A	N/A	+110	+67	(47)

Nomenclature: See separate page